PART A — QUESTIONS

AF25-001
Question:
Solve: $27x - 20 = 142$
A) 5
B) 6
C) 7
D) 8

AF25-002
Question:
If $25n + 15 = 190$, what is n?
A) 5
B) 6
C) 7
D) 8

AF25-003
Question:
Simplify: $12(2x - 5) - 9x$
A) $15x - 60$
B) $24x - 60$
C) $15x - 5$
D) $33x - 60$

AF25-004
Question:
A company charges $60 per shipment plus a $35 processing fee. Which expression represents the total charge for s shipments?
A) $60s + 35$
B) $35s + 60$
C) $95s$
D) $60 - 35s$

AF25-005
Question:
If $f(x) = 22x - 85$, what is $f(8)$?
A) 81
B) 91
C) 176
D) 261

AF25-006
Question:
Which ordered pair satisfies the equation $y = 20x - 11$?
A) (1, 10)
B) (2, 29)
C) (3, 49)
D) (4, 70)

AF25-007
Question:
Solve: $32(x - 8) = 224$
A) 13
B) 14
C) 15
D) 16

AF25-008
Question:
A pattern begins: 20, 44, 68, 92, ... What is the next number?
A) 114
B) 115
C) 116
D) 117

AF25-009
Question:
Solve: $x/65 = 3$
A) 68
B) 130
C) 195
D) 260

AF25-010
Question:
A line has a slope of -22 and crosses the y-axis at 19. Which equation represents the line?
A) $y = 19x - 22$
B) $y = -22x + 19$
C) $y = 22x + 19$
D) $y = -19x + 22$

AF25-011
Question:
If 8 containers hold 376 parts, how many parts are in 6 containers at the same rate?
A) 264
B) 276
C) 282
D) 288

AF25-012
Question:
Solve: $600 - 52x = 392$
A) 3
B) 4
C) 5
D) 6

AF25-013
Question:
Which expression is equivalent to $160x - 74x - 25$?

A) 86x - 25
B) 234x - 25
C) 86x + 25
D) 160x - 99

AF25-014
Question:
A stockroom starts with 3,200 items. Each work order uses 200 items. Which equation gives the number of items I left after w work orders?
A) I = 3,200w - 200
B) I = 200w + 3,200
C) I = 3,200 - 200w
D) I = 200w - 3,200

AF25-015
Question:
If y = -40x + 700, what is y when x = 16?
A) 40
B) 60
C) 640
D) 1,340

AF25-016
Question:
Solve: 44x - 132 = 32x + 48
A) 13
B) 14
C) 15
D) 16

AF25-017
Question:
Which value of x makes 28x - 15 less than 405?
A) 14
B) 15
C) 16
D) 17

AF25-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: -14, 11, 36, 61

What is the rule?
A) y = 25x - 14
B) y = 14x + 25
C) y = -14x + 25
D) y = 25x + 14

AF25-019
Question:
Solve: $28(x + 6) = 25x + 210$
A) 12
B) 13
C) 14
D) 15

AF25-020
Question:
If $a = -20$ and $b = 17$, what is $5a + 6b$?
A) -202
B) -2
C) 2
D) 202

AF25-021
Question:
Which point is located in Quadrant II?
A) (21, 12)
B) (-21, 12)
C) (-21, -12)
D) (21, -12)

AF25-022
Question:
A worker sorts 360 items in 18 minutes. At the same rate, how many items can the worker sort in 7 minutes?
A) 120
B) 130
C) 140
D) 150

AF25-023
Question:
Solve for r: $J = 50r - 150$
A) $r = J + 150$
B) $r = (J + 150)/50$
C) $r = 50(J + 150)$
D) $r = J/50 - 150$

AF25-024
Question:
What is the slope of a line that goes through (14, 30) and (20, 144)?
A) 17
B) 18
C) 19
D) 114

AF25-025
Question:

Simplify: $120x + 34 - 109x - 60$
A) $11x - 26$
B) $229x - 26$
C) $11x + 94$
D) $229x + 94$

AF25-026
Question:
If $g(x) = x^2 + 9x$, what is $g(15)$?
A) 225
B) 270
C) 360
D) 450

AF25-027
Question:
A sequence follows the rule “add 16, then multiply by 5.” If the first term is 8, what is the second term?
A) 24
B) 80
C) 120
D) 160

AF25-028
Question:
Solve: $32(x - 7) + 56 = 344$
A) 14
B) 15
C) 16
D) 17

AF25-029
Question:
A line crosses the y-axis at 24 and falls 15 units for every 1 unit to the right. Which equation represents the line?
A) $y = 15x + 24$
B) $y = -15x + 24$
C) $y = 24x - 15$
D) $y = -24x + 15$

AF25-030
Question:
If $44/66 = x/198$, what is x?
A) 120
B) 126
C) 132
D) 138

AF25-031
Question:
Solve: $-75x + 375 = -750$

A) -15
B) -10
C) 10
D) 15

AF25-032
Question:
A warehouse starts with 12,000 fasteners. Each project uses 600 fasteners. Which expression gives the number of fasteners left after p projects?
A) 12,000 + 600p
B) 12,000 - 600p
C) 600p - 12,000
D) 12,000 ÷ 600p

AF25-033
Question:
Which equation has x = 36 as its solution?
A) 2x + 25 = 95
B) 5x - 70 = 110
C) x/18 + 8 = 11
D) 3x - 24 = 82

PART B — ANSWER KEY + EXPLANATIONS

AF25-001 — Correct: B — Explanation:
Solve 27x - 20 = 142. Add 20 to both sides: 27x = 162. Divide by 27: x = 6. The most common mistake is subtracting 20 instead of adding 20.

AF25-002 — Correct: C — Explanation:
Start with 25n + 15 = 190. Subtract 15 from both sides: 25n = 175. Divide by 25: n = 7. A common mistake is stopping at 175 and forgetting to divide by 25.

AF25-003 — Correct: A — Explanation:
Distribute first: 12(2x - 5) = 24x - 60. Then subtract 9x: 24x - 9x - 60 = 15x - 60. A common mistake is forgetting to multiply -5 by 12.

AF25-004 — Correct: A — Explanation:
The company charges $60 for each shipment, so that part is 60s. The $35 processing fee is added one time. The total charge is 60s + 35. A common mistake is treating the processing fee as the per-shipment charge.

AF25-005 — Correct: B — Explanation:
Substitute 8 for x: f(8) = 22(8) - 85 = 176 - 85 = 91. A common mistake is adding 85 instead of subtracting it.

AF25-006 — Correct: C — Explanation:
Test the choices in y = 20x - 11. For (3, 49), 20(3) - 11 = 60 - 11 = 49, which matches the y-value. A common mistake is choosing a point that is close but does not make the equation true.

AF25-007 — Correct: C — Explanation:
Solve 32(x - 8) = 224. Divide both sides by 32: x - 8 = 7. Add 8: x = 15. A common mistake is adding 8 before undoing the multiplication by 32.

AF25-008 — Correct: C — Explanation:
The pattern increases by 24 each time: 20, 44, 68, 92. Add 24 to 92: 116. A common mistake is assuming the increase is 22 or 23 without checking the differences.

AF25-009 — Correct: C — Explanation:
x/65 = 3 means x divided by 65 equals 3. Multiply both sides by 65: x = 195. A common mistake is adding 65 and 3 instead of multiplying.

AF25-010 — Correct: B — Explanation:
Slope-intercept form is y = mx + b. The slope is -22 and the y-intercept is 19, so the equation is y = -22x + 19. A common mistake is switching the slope and y-intercept.

AF25-011 — Correct: C — Explanation:
Find the unit rate: 376 parts ÷ 8 containers = 47 parts per container. For 6 containers: 47 × 6 = 282. A common mistake is multiplying 376 by 6 without finding the rate first.

AF25-012 — Correct: B — Explanation:
Solve 600 - 52x = 392. Subtract 600 from both sides: -52x = -208. Divide by -52: x = 4. A common mistake is losing the negative sign after subtracting 600.

AF25-013 — Correct: A — Explanation:
Combine like terms: 160x - 74x = 86x. The constant -25 stays the same. The expression becomes 86x - 25. A common mistake is combining the constant with the variable terms.

AF25-014 — Correct: C — Explanation:
The stockroom starts with 3,200 items. Each work order uses 200 items, so w work orders use 200w items. The number left is I = 3,200 - 200w. A common mistake is adding 200w even though items are being used.

AF25-015 — Correct: B — Explanation:
Substitute x = 16 into y = -40x + 700: y = -40(16) + 700 = -640 + 700 = 60. A common mistake is treating -40(16) as positive 640.

AF25-016 — Correct: C — Explanation:
Solve 44x - 132 = 32x + 48. Subtract 32x: 12x - 132 = 48. Add 132: 12x = 180. Divide by 12: x = 15. A common mistake is adding 132 to only one side.

AF25-017 — Correct: A — Explanation:
Solve 28x - 15 < 405. Add 15: 28x < 420. Divide by 28: x < 15. The only listed value less than 15 is 14. A common mistake is choosing 15, but 28(15) - 15 = 405, not less than 405.

AF25-018 — Correct: A — Explanation:
The y-values increase by 25 each time x increases by 1, so the slope is 25. When x = 0, y = -14, so the y-intercept is -14. The rule is y = 25x - 14. A common mistake is using +14 instead of -14.

AF25-019 — Correct: C — Explanation:
Distribute: $28x + 168 = 25x + 210$. Subtract 25x: $3x + 168 = 210$. Subtract 168: $3x = 42$. Divide by 3: $x = 14$. A common mistake is forgetting to multiply 6 by 28.

AF25-020 — Correct: C — Explanation:
Substitute $a = -20$ and $b = 17$: $5a + 6b = 5(-20) + 6(17) = -100 + 102 = 2$. A common mistake is treating 5(-20) as positive 100.

AF25-021 — Correct: B — Explanation:
Quadrant II has a negative x-coordinate and a positive y-coordinate. The point (-21, 12) fits this description. A common mistake is choosing (21, -12), which is Quadrant IV.

AF25-022 — Correct: C — Explanation:
360 items in 18 minutes means $360 \div 18 = 20$ items per minute. In 7 minutes: $20 \times 7 = 140$. A common mistake is multiplying 360 by 7 without finding the rate first.

AF25-023 — Correct: B — Explanation:
Start with $J = 50r - 150$. Add 150 to both sides: $J + 150 = 50r$. Divide by 50: $r = (J + 150)/50$. A common mistake is dividing J by 50 first and then adding 150.

AF25-024 — Correct: C — Explanation:
Slope equals change in y divided by change in x. From (14, 30) to (20, 144), y increases by 114 and x increases by 6. The slope is $114 \div 6 = 19$. A common mistake is using 114 as the slope without dividing by the change in x.

AF25-025 — Correct: A — Explanation:
Combine like terms: $120x - 109x = 11x$, and $34 - 60 = -26$. The simplified expression is $11x - 26$. A common mistake is writing $11x + 26$ because the negative sign before 60 is missed.

AF25-026 — Correct: C — Explanation:
Substitute 15 for x: $g(15) = 15^2 + 9(15) = 225 + 135 = 360$. A common mistake is calculating 15^2 as 30 instead of 225.

AF25-027 — Correct: C — Explanation:
Start with 8. Add 16: 24. Then multiply by 5: 120. A common mistake is multiplying first and then adding 16.

AF25-028 — Correct: C — Explanation:
Solve $32(x - 7) + 56 = 344$. Distribute: $32x - 224 + 56 = 344$. Combine: $32x - 168 = 344$. Add 168: $32x = 512$. Divide by 32: $x = 16$. A common mistake is forgetting to combine -224 and +56.

AF25-029 — Correct: B — Explanation:
“Falls 15 units for every 1 unit to the right” means the slope is -15. The line crosses the y-axis at 24, so the equation is $y = -15x + 24$. A common mistake is using a positive slope because the number 15 is given.

AF25-030 — Correct: C — Explanation:
Use the proportion $44/66 = x/198$. Since $66 \times 3 = 198$, multiply 44 by 3: $x = 132$. A common mistake is adding 132 to the numerator because 66 becomes 198.

AF25-031 — Correct: D — Explanation:
Solve -75x + 375 = -750. Subtract 375 from both sides: -75x = -1,125. Divide by -75: x = 15. A common mistake is giving -15 because both sides contain negative numbers.

AF25-032 — Correct: B — Explanation:
The warehouse starts with 12,000 fasteners. Each project uses 600 fasteners, so p projects use 600p fasteners. The number left is 12,000 - 600p. A common mistake is adding 600p even though fasteners are being used.

AF25-033 — Correct: B — Explanation:
Test x = 36. In choice B, 5(36) - 70 = 180 - 70 = 110, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.